

Seismic Surface-Related Multiples Suppression Based on SAGAN

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Abstract—Suppressing multiples from seismic records is necessary to improve imaging quality. Deep neural networks (DNNs) can automatically mine features from data. Once a network is successfully trained, it can process data with extremely high efficiency. In this letter, a generative adversarial network (GAN) framework is proposed to remove surface-related multiples in both synthetic and field datasets, where the generator is U-Net with Markov discriminator. Adding self-attention (SA) blocks to GAN improves processing precision. Improved signal noise ratio (SNR), and accurate reverse time migration (RTM) images implemented by network's outputs of synthetic datasets, jointly support that this network is effective on surface-related multiple suppression. Based on the results from field application, deep learning method in this letter is comparable to conventional adaptive surface-related multiple elimination (SRME) method but time-saving. By constructing an end-to-end workflow for seismic surface-related multiples suppression, small batches dataset can be used to train the network, and large batches of datasets can be processed accurately and efficiently.

Index Terms—Data processing, deep neural network (DNN), generative adversarial network (GAN), self-attention (SA), surface-related multiple.

I. INTRODUCTION

FOR reflected seismic waves, if they encounter an interface with strong impedance difference, secondary downward energy is formed. This energy represents the waves that have reflected two or more times off the former interface and is known as a “multiple.” For great impedance differences between air, water, and sedimentary seabed, multiples are very common in on- and off-shore seismic surveys of the geological structures of oil and gas reservoirs. Surface-related multiples will be reflected off the top interface (ground or sea surface) once at least, as shown in Fig. 1. However, the basic assumption of most seismic migration imaging algorithms is that all energy reflects only once underground. Therefore, multiples can lead to the misinterpretation of geological structures,

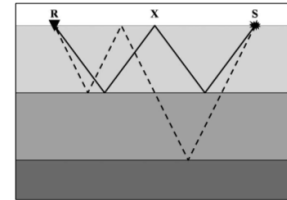


Fig. 1. Two possible surface-related multiple wave-paths with the same shot and receiver.

which often results in them being regarded as noise and are required to be removed from seismic data [1].

With practical scenarios, data-driven methods would not be limited by the selection of fussy thresholds or the detailed parameters of seismic models. Surface-related multiple elimination (SRME) driven by seismic data have been the standard method about surface-related multiple removal [2]. It works in two steps, predicting multiples based on feedback-iterative spatial convolution and adaptive subtracting from original seismic records due to the mismatched amplitude about multiples [3]. But it has high requirements for densely sampled data and high degree of computation for adaptive subtraction. Many researchers desire to tackle with these problems. Fomel [4] adopt regularized optimization to non-stationary matching filtering. A new fast local primary-and-multiple orthogonalization framework is proposed to extract leaky multiples and improve efficiency [5]. Zhang *et al.* [6] utilized orthogonalization for leaked internal multiple crosstalk estimation and attenuation on full-wavefield migrated images. Qu *et al.* proposed a deep learning-based method to reconstruct missing near-offset and remove multiple on field data [7].

In contrast with conventional techniques, deep learning (DL) can mine goal features from datasets automatically based on statistics and optimality by nonlinear mapping multilayer networks. It has so great power on generalization that we can train a DL model by label few data manually to process big-body seismic data. The efficiency would be improved, and time cost would be reduced as well as human cost. U-Net [8], which is based on convolutional neural network (CNN), is famous for outstanding achievements on medical images segment. Actually, these methods are common in both geophysical data processing and medical image processing because of the similarity with the two fields. Residual U-Net reconstructs sparsely sampled seismic data [9]. Deblending of seismic data based on U-Net [10]. CNN reduces the parameters of the network using local connections; however, this it usually

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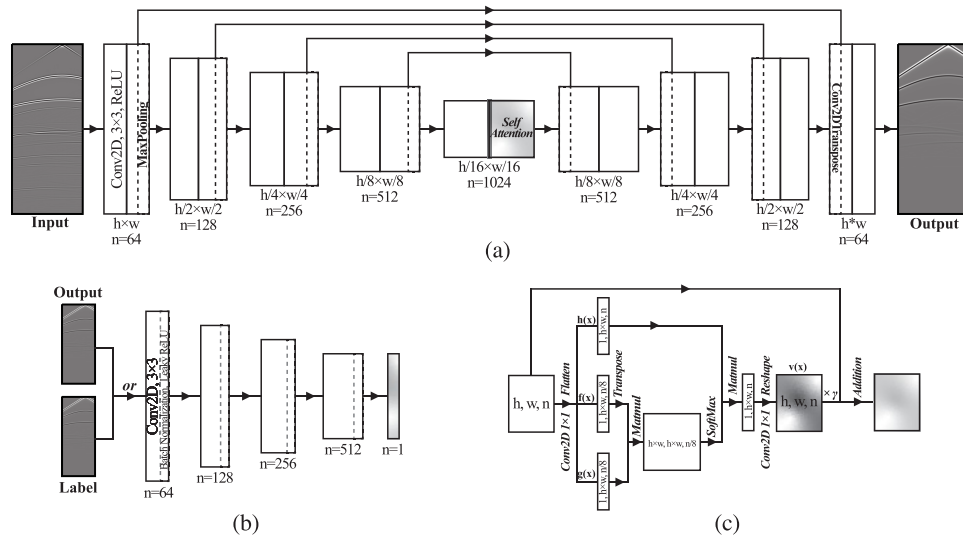


Fig. 2. SAGAN architecture in this letter. It includes three parts (a) U-Net generator, (b) Markov discriminator, and (c) SA block.

leads to missing global information. Self-attention (SA) [11] can assign corresponding weights to feature maps to allow redundant information to be ignored. SAGAN [12] can use remote information to ensure consistent improvements in the quality of the images generated.

Goodfellow [13] proposed a generative adversarial network (GAN) that followed the principles of probability and game theory. GAN is one of DL frameworks composed of two or more deep neural networks (DNNs). The networks oppose, encourage, and improve each other until finally, the outputs became as close as they could be to the label. Wasserstein GAN [14] with gradient penalty (WGAN-GP [15]) introduced Wasserstein distance as differences measurement between two distributions and gradient penalty to avoid unstable training in traditional GAN. In recent years, GANs with different structures have been widely used to augment, process, and interpret geophysical data. Pixel2Pixel is used to eliminate surface-related multiples [16]. Combined with hyperbolic Radon transformation, this separates the multiples and primaries in common shot gathers [17]. WGAN-GP had been applied to impedance inversion [18].

This letter combines GAN with U-Net, WGAN-GP, and SA to suppress seismic surface-related multiples in shot domain. As part of the need to establish datasets corresponding to our goal, DNNs automatically learn the features of interests from the data and the network outputs the expected result. If the relationship mapped between input and label by seismic records with and without surface-related multiples, the outputs no longer need to be adaptively subtracted. Datasets would be divided into two parts for training and test, respectively, by 2:8 at regular intervals. It aims to prove that network can learn from small portion of data processed by specialist, and process big-body data with less time and reduced computation cost after generalization. To validate our network is effective and SA block can improve precision, application on synthetic datasets creating by horizontal layered velocity models and corresponding reverse time migration (RTM) images are given. RTM images could show the influence led by surface-related

multiples in seismic imaging. Concrete analysis about results that processing field data by SAGAN is following. Signal noise ratio (SNR) is chosen to evaluate network's outputs.

II. NETWORK STRUCTURES AND TRAINING STRATEGY

A. Network Architecture

The U-Net as generator, shown in Fig. 2(a), was composed of three parts—an encoder, decoder, and shortcut connections—and was based on convolutional hidden layers. The encoder compressed the input data and obtained a sparse indication, the decoder reconstructed the data by restoring the encoder output, and the shortcut connections fused shallow and deep information and thus avoided the vanishing gradient. In the encoding, five groups of convolutions were implemented by the number of channels that doubled in a corresponding manner, i.e., from 1 channel to 64, 128, 256, 512, and 1024 successively. Each convolution group was composed of two 3×3 filters and the size of the feature maps remained unchanged. The max-pooling after each group convolution was executed, and the length and width of the feature maps were halved after pooling. In this process, the features of the inputs are extracted by a reduction in dimensionality, and sparse feature maps are obtained. Encoder and decoder linked by a SA block, shown in Fig. 2(c). In the decoding, the sparse feature maps are first deconvoluted so that their length and width are doubled. It is also a U-Net rule that before deconvolution can occur, the decoding feature maps are concatenated with the feature maps of the corresponding encoding layers. Similarly, four groups of deconvolutions were implemented successively by the corresponding number of halved channels, from 1024 to 512, 256, 128, and 64. After each deconvolution group, deconvolution was performed until the length and width of the results were consistent with those of the inputs. Finally, a single-channel deconvolution was used to fuse the feature maps with 64 channels. This ensured that the size and dimension of the outputs of the U-Net

were consistent with the inputs. The features of interest were restored after decoding.

Discriminator is shown in Fig. 2(b). It was derived from Pixel2Pixel and is called a Markov discriminator. It consisted of four convolutional layers with 4×4 filters, two strides, and 64, 128, 256, and 512. Batch normalization and leaky rectified linear unit activation were conducted after each convolution. Then, followed by a SA block. Finally, the discriminating result is given by a single channel 4×4 filter with one stride.

The feature maps from previous hidden layers $\mathbf{x} \in \mathbb{R}^{N \times C}$ are first transformed into $f(\mathbf{x})$, $g(\mathbf{x})$ and $h(\mathbf{x})$ by convolution to obtain the attention [12], where N is the size of feature maps and C is the number of channels, $N = h \times w$, $C = n$. $\beta_{i,j} = ((\exp(s_{ij})) / (\sum_{i=1}^N \exp(s_{ij})))$, where $s_{ij} = f(\mathbf{x}_i)^T g(\mathbf{x}_j)$. $\beta_{i,j}$ indicates the extent to which the model attends to the i th location when synthesizing the j th region. The output is $\mathbf{o} = (\mathbf{o}_1, \mathbf{o}_2, \dots, \mathbf{o}_j, \dots, \mathbf{o}_N) \in \mathbb{R}^{N \times C}$, where $\mathbf{o}_j = v(\sum_{i=1}^N \beta_{i,j} h(\mathbf{x}_i))$. The output of SA is $y_i = \gamma \mathbf{o}_i + \mathbf{x}_i$.

B. Training Strategy

The WGAN-GP loss function is defined by (1) [15], $\mathbf{x}$ and $\mathbf{y}$ are inputs and labels (gather with and without surface-related multiples), $\tilde{\mathbf{x}}$ is predicted by generator. $\hat{\mathbf{x}}$ is defined by (2), sampled points along the straight lines between $\mathbf{y}$ and $\tilde{\mathbf{x}}$, where ϵ is a random number $\epsilon \sim U[0, 1]$, λ is a coefficient of gradient penalty. The gradient penalty would be added on discriminator's loss function. L_2 regularization is applied to generator's loss function as shown in (3)

$$L = -\mathbb{E}_{\tilde{\mathbf{x}} \sim \mathbb{P}_g} [D(\tilde{\mathbf{x}})] + \mathbb{E}_{\mathbf{y} \sim \mathbb{P}_r} [D(\mathbf{y})] + \lambda \mathbb{E}_{\hat{\mathbf{x}} \sim \mathbb{P}_{\tilde{\mathbf{x}}}} [(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2] \quad (1)$$

$$\hat{\mathbf{x}} = \epsilon \mathbf{y} + (1 - \epsilon) \tilde{\mathbf{x}} \quad (2)$$

$$L_{\text{MSE}} = \|\mathbf{y} - \tilde{\mathbf{x}}\|_2^2. \quad (3)$$

Generator's loss function is formulated as follows:

$$L_g = -\mathbb{E}_{\tilde{\mathbf{x}} \sim \mathbb{P}_g} [D(\tilde{\mathbf{x}})] + \lambda_1 \mathbb{E}_{\tilde{\mathbf{x}} \sim \mathbb{P}_g, \mathbf{y} \sim \mathbb{P}_r} [\|\mathbf{y} - \tilde{\mathbf{x}}\|_2^2]. \quad (4)$$

Discriminator's loss function is formulated as follows:

$$L_d = -\mathbb{E}_{\tilde{\mathbf{x}} \sim \mathbb{P}_g} [D(\tilde{\mathbf{x}})] + \mathbb{E}_{\mathbf{y} \sim \mathbb{P}_r} [D(\mathbf{y})] + \lambda_2 \mathbb{E}_{\hat{\mathbf{x}} \sim \mathbb{P}_{\tilde{\mathbf{x}}}} [(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2] \quad (5)$$

where λ_1, λ_2 are coefficients for regulation and gradient penalty, respectively. Meng *et al.* [18] gave a value reference, $\lambda_1 = 1000$, $\lambda_2 = 10$. Adam algorithm [19] is used in the training with a batch size of 1 and learning rate of 0.001.

Data preprocessing is very important for training neural networks because they learn the implicit distribution of data [19]; therefore, preprocessing can prevent overfitting. This letter used Z-score standardization was used to preprocess the data as shown in the following:

$$\mathbf{x} = \frac{\mathbf{x}_0 - \mu}{\sigma}. \quad (6)$$

Considering multiple is also a kind of seismic signals in fact and owns global and continuous features in phase and amplitude, this letter is prone to input per trace gather without patch and output corresponding trace gather after multiples removal.

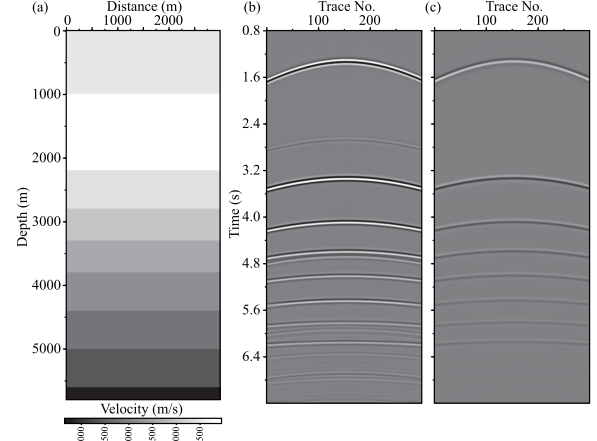


Fig. 3. Synthetic velocity models and seismic datasets. (a) Velocity. (b) One shot data with surface-related multiples as network input. (c) Corresponding shot data without surface-related multiples as network label.

The pairs of data will be divided into two parts by 2:8 for training and test. Training datasets would be also divided into two parts by 2:8 for training and validating network's parameters and hyper-parameters. For applications on both synthetic and field datasets, initially, 300 training epoch is set. The optimal network parameters judged by SNR in validation will be saved automatically. SNR is defined by power of signal and noise as follows [20]:

$$\text{SNR} = 10 \log_{10} \left(\frac{\|\mathbf{y}\|_F^2}{\|\mathbf{y} - \tilde{\mathbf{x}}\|_F^2} \right). \quad (7)$$

III. APPLICATIONS ON SYNTHETIC DATASETS

A. Datasets

Forward modeling based on the wave equation was conducted using the finite difference method to obtain synthetic datasets. In all 15 velocity models [20], line distance is 3000 m, and depth is 5800 m. A marine surface layer with a depth of 1000 m and wave velocity of 1500 m/s is placed above the seabed to simulate a marine seismic survey. When the free boundary was set, the synthetic seismic common shot gathers had surface-related multiples that were regarded as input datasets. When the absorbing boundary was set, the synthetic gathers did not have surface-related multiples and were regarded as label datasets. We set 100 shots and 300 receivers at equivalent interval for each velocity model, and receivers record signals per 8 ms. Each common shot gather (i.e., input) is 300 traces and 900 time points sized. One of velocity models and its corresponding seismic data are shown in Fig. 3. For per velocity model, there are 20 shots for training and the rest 80 shots for testing.

B. Training

In this letter, each training epoch would consume 84.60 s for 300 iterations. Based on Figs. 4(a) and 5, epoch average SNR is gradually improved, and surface-related multiples [locate in white arrows in Fig. 5(a)] are suppressed. The maximum of epoch average SNR is 54.17 dB at 266 epoch which is 0.71 dB before training. Although network also did well at

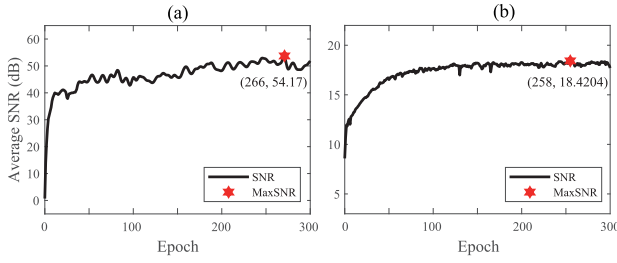


Fig. 4. Epoch average SNR curve in training. (a) Synthetic. (b) Field.

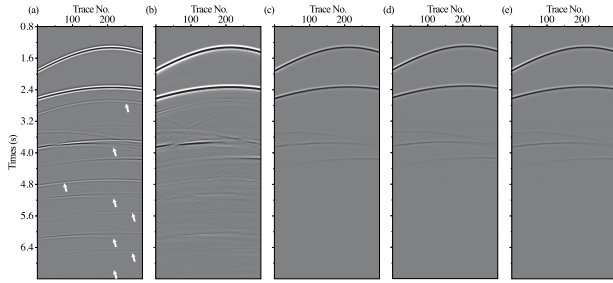


Fig. 5. Sample of output in validation at increasing epoch shows network's convergence. (a) Input data, SNR = 0.31 dB. (b) Validation at epoch = 1, SNR = 25.15 dB. (c) Validation at epoch = 100, SNR = 45.13 dB. (d) Validation at epoch = 266, SNR = 54.01 dB. (e) Label data.

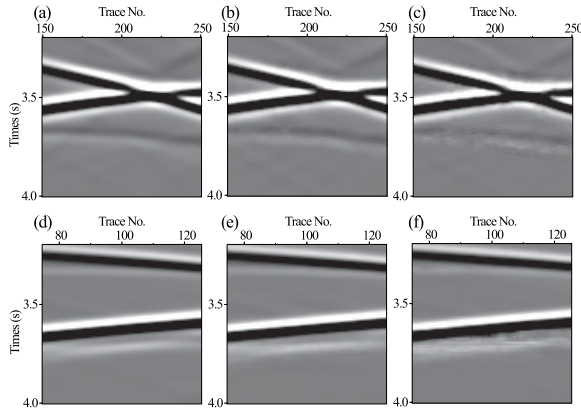


Fig. 6. SA blocks contribute for test result. (a) and (d) Label. (b) and (e) Result zoomed-in-view with SA blocks. (c) and (f) Result without SA block.

epoch 100 [in Fig. 5(c)], validation keeps better continuity for in-phase axes of primaries at epoch 266.

C. Test Results

There are 1200 test pairs that are 0.45 dB originally in average, and their SNR is improved to 50.17 dB after processing by our network for 62.00 s. To verify the contribution of SA blocks for our network, two local seismic gathers are given in Fig. 6. SA blocks make results more accurate and precise. To intuitively verify and demonstrate the correctness and effectiveness of output shot gathers, RTM was conducted by inputs, predicts, and labels, RTM images are list in Fig. 7. Imaging by inputs is prone to turn out spurious interfaces as marked by the higher arrow in Fig. 7(a), and these deeper layers cannot be recognized as marked by the lower arrow.

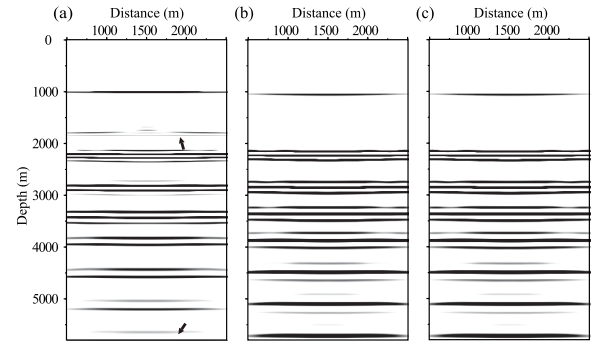


Fig. 7. RTM images by (a) input shot gathers, (b) output shot gathers after network processing (a), and (c) label shot gathers.

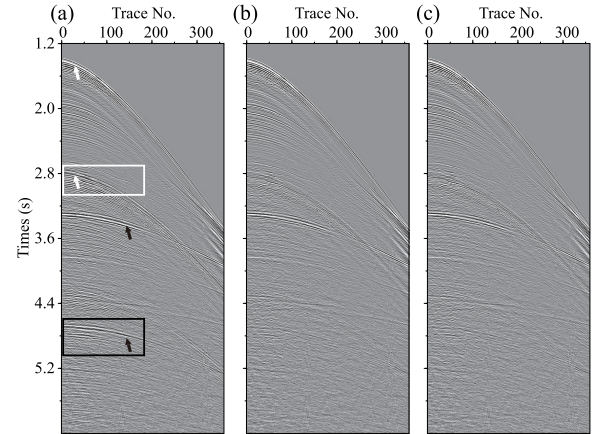


Fig. 8. Result of testing field data. (a) Input, two groups of primaries and their corresponding surface-related multiples are marked by white and black arrows, respectively. Higher one marks primaries, the other one marks multiples. (b) Network output. (c) Label.

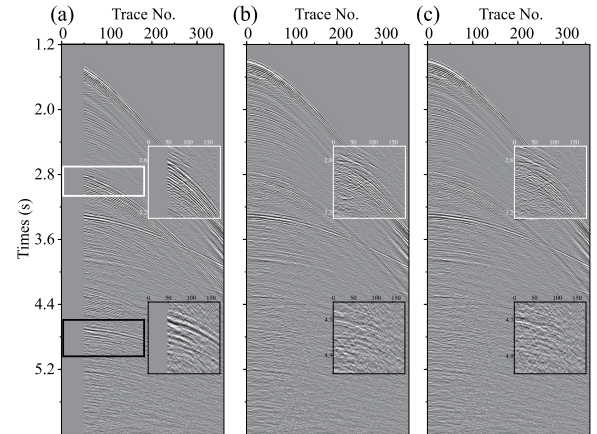


Fig. 9. Result of near-offset missing field data. (a) Input, SNR = 3.77 dB. (b) Network output, SNR = 13.44 dB. (c) Label.

IV. APPLICATIONS ON FIELD DATASETS

A. Dataset

Field datasets are common shot gathers and consistent with synthetic datasets. There are 600 pairs of gather in order of source location, and per gather (i.e., input) is 360 traces and 1201 time points sized, as Figs. 8 and 9. Keeping split ratio by 2:8 in this letter, dividing into 120 pairs for training and 480 for test at same source location interval.

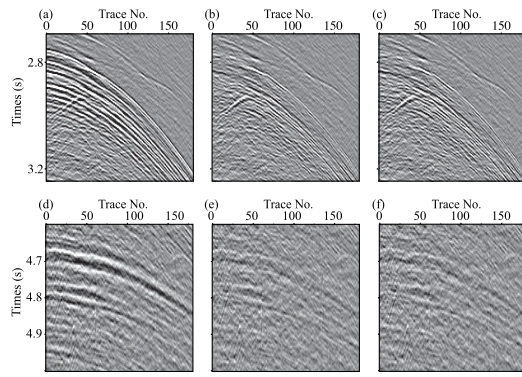


Fig. 10. Boxes zoomed-in-view of surface-related multiples in Fig. 8. (a)–(c) White box. (d)–(f) Black box. (a) and (d) Input. (b) and (e) Network output. (c) and (f) Result by adaptive SRME.

B. Training and Test Results

Each epoch consumed 55.44 s in training. For processing 480 test pairs, our method only took 39.55 s. Fig. 4(b) gives SNR curves in above two processes. In test, the maximum SNR of network's output is 18.66 dB. Fig. 8 is a sample from test dataset. Two groups of primaries and their corresponding surface-related multiples are marked by white and black arrows, respectively. White box is zoomed and displayed in Fig. 10(a)–(c), and black one in Fig. 10(d)–(f), to focus on multiples. Overall, prediction is close to label, network can recognize those regions of multiples and suppress them. In Fig. 10, energy of surface-related multiples is suppressed obviously, and that of primaries is retained well. Fig. 10(e) shows that our network in this letter can remove these multiples which are not as strong as those in Fig. 10(b).

In addition, this network is applied for near-offset missing cases, and 50 traces missing in this letter. After training in the same strategy, one of testing results is shown in Fig. 9. It shows the phase and amplitude is both recovered.

V. CONCLUSION

This letter applied SAGAN on both synthetic and field datasets. Datasets are divided into two portions by 2:8 ratio for training and test, respectively. After training with small batches dataset, the generalization is successful, and large batches of dataset can be processed automatically and effectively. In synthetic experiment, outputs from network, SNR analysis, and RTM results all show that surface-related multiples from horizontal layered models can be suppressed accurately. The SA module improves network performance. Experiments on the field datasets show that the proposed network can correctly identify and suppress multiples, the primary waves can be maintained, and the overall effect is equivalent to SRME. Rather SRME and adaptive subtraction, our method is time-saving because only tens of milliseconds consumed for per seismic data slice predicted by trained network. The efficiency of data processing was greatly improved. For near-offset

missing cases, it's effective for both interpolation and multiple removal. Direct mapping from with to without surface-related multiples data gives an end-to-end solution for seismic data processing.

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